

Open-Source EDA and PDKs for Teaching and Research in the PULP Group

DAC 2026, Open-Source EDA, Birds of a Feather

Arpan Suravi Prasad	prasadar@iis.ee.ethz.ch
Philippe Sauter	psauter@iis.ee.ethz.ch
Thomas Benz	tbenz@iis.ee.ethz.ch
Paul Scheffler	paulsc@iis.ee.ethz.ch
Gamze Islamoglu	gislamoglu@iis.ee.ethz.ch
Frank K. Gürkaynak	kgf@iis.ee.ethz.ch
Luca Benini	lbenini@iis.ee.ethz.ch

PULP Platform

Open Source Hardware, the way it should be!



pulp-platform.org

[@pulp_platform](https://twitter.com/pulp_platform)

[company/pulp-platform](https://www.linkedin.com/company/pulp-platform)

[youtube.com/pulp_platform](https://www.youtube.com/pulp_platform)

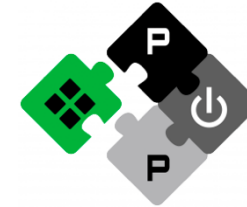


PULP platform has a long history in open source HW



- **PULP (Parallel Ultra Low Power) Platform started in 2013**

- With the appointment of Luca Benini as a professor to ETH Zurich while keeping his professorship at the University of Bologna



ETH zürich



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

- **From the beginning, the group was active in open source hardware**

- First open source project was PULPino, released in 2016



- **Starting in 2015, the group started developing RISC-V-based systems**

- CVA6, CVE32E40P, Ibex/CVE2 all started their life as PULP platform projects



**The PULP team is
about 100 people
Both in CH and IT**

As part of our adventure we have designed **86** ASICs



There are **three connected** aspects of open source HW



Design

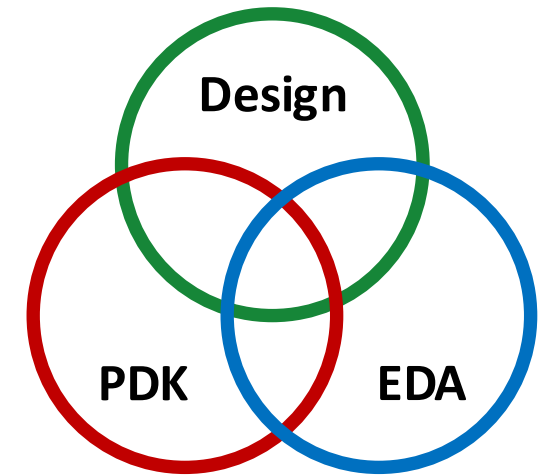
- RTL / HDL descriptions (quite common)
- Schematics / Physical Design (may have dependencies to technology information)

Tools (EDA)

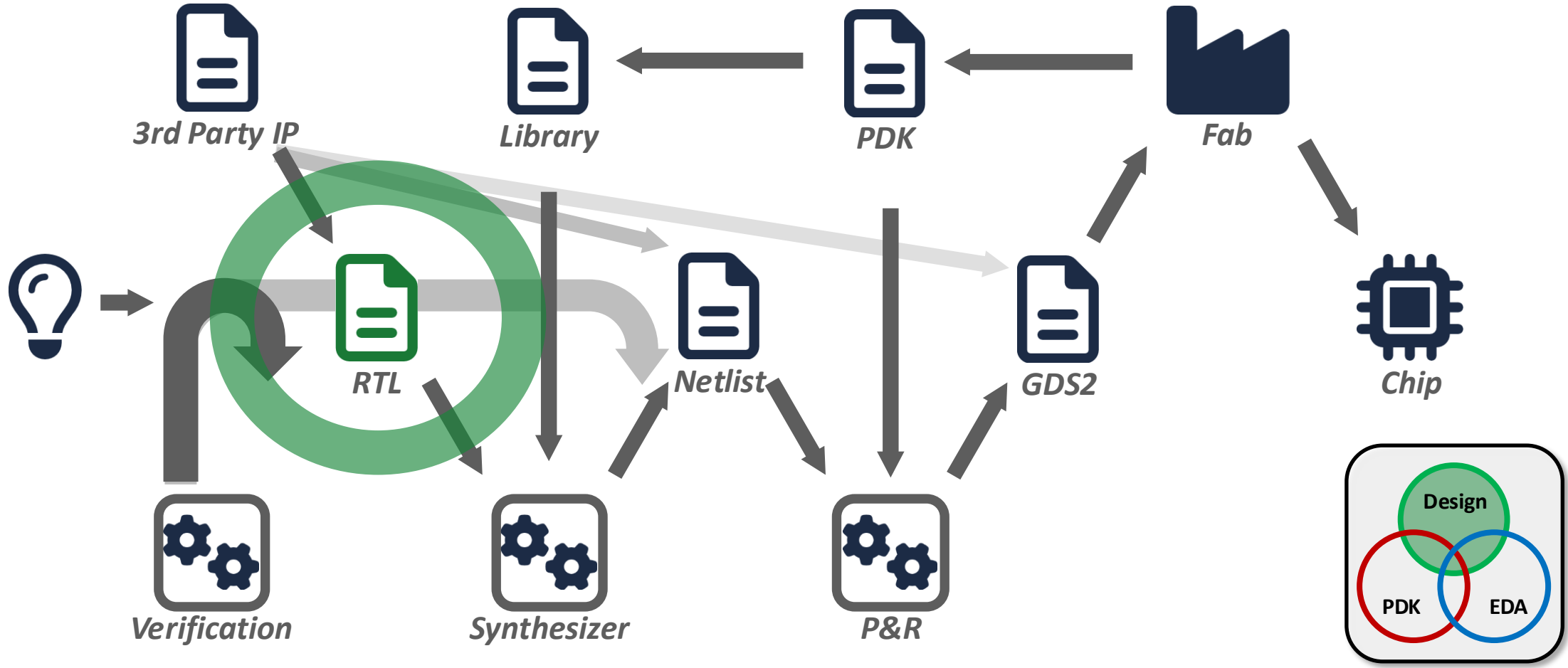
- Front-end tools (Synthesis)
- Back-end tools (Placement and Routing)
- Verification tools (Simulation)

Manufacturing (PDK)

- Design rules for manufacturing (separation, minimum width of metals)
- Layer stack information for parasitics (thickness, dielectric constants..)
- Device models (SPICE parameters) for simulation



Most of open source hardware today is at **design** level



All PULP contributions are available as open-source HW



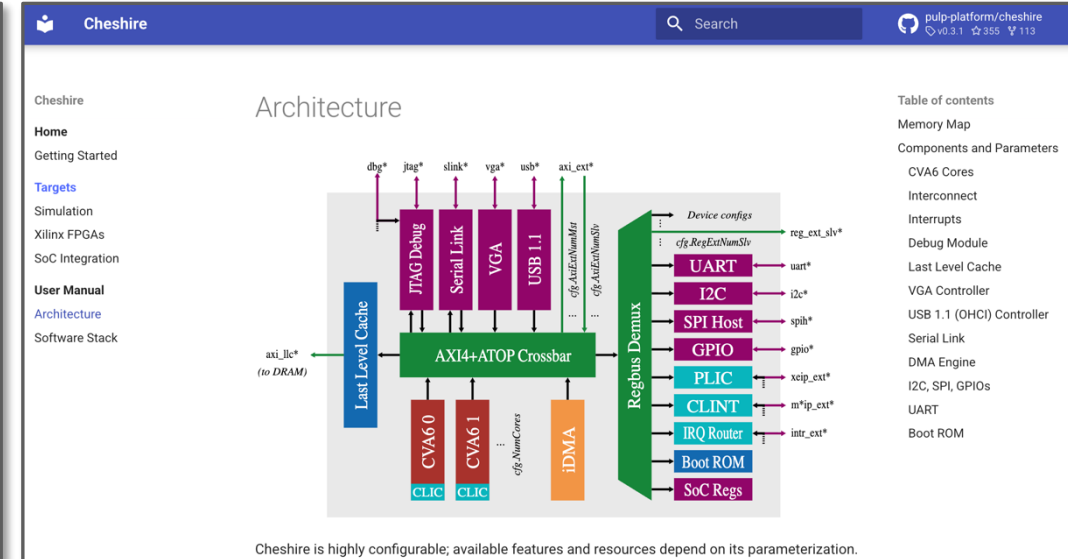
- All our development is on GitHub using a permissive license
 - HDL source code, testbenches, software development kit, virtual platform

<https://github.com/pulp-platform>



- Allows anyone to use, change, and make products without restrictions.

The screenshot shows the GitHub repository for 'pulp-platform'. The repository is public and has 326 repositories, 2 projects, and 13 people. It features a 'Deeploy' button and a description: 'DNN Compiler for Heterogeneous SoCs'. The repository is written in Python, has 71 stars, and 45 forks. Other repositories listed include 'carfield' (a mixed-criticality platform), 'cheshire' (a minimal Linux-capable 64-bit RISC-V SoC), and 'snitch_cluster' (an energy-efficient RISC-V floating-point compute cluster).

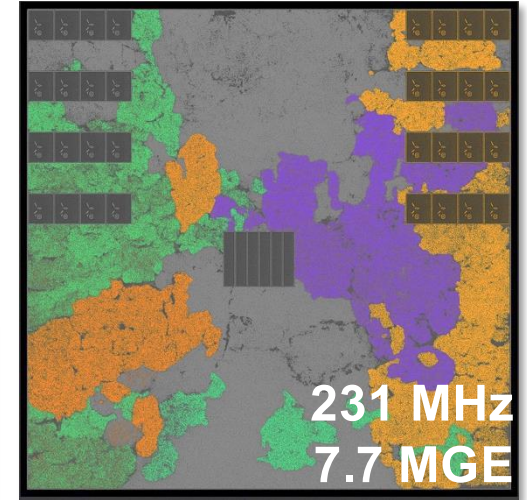


We contribute to **Open EDA** for better designs

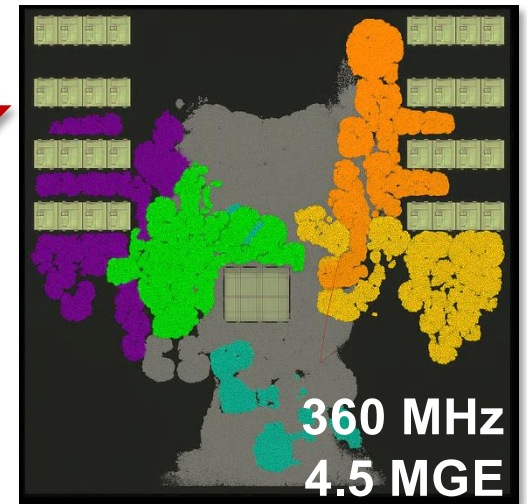


- **Established open-source EDA tools that deliver real silicon**
 - OpenROAD, Yosys, ABC and many others
- **PULP team is working with developers to close PPA gaps**
 - There is some gap, but it is smaller than what many expect
 - Very good results for 180-130nm designs
 - More work & experience needed for more modern nodes
- **Able to influence the "inner loop" for AI-assisted methods**
 - Huge potential for innovation and development

Baseline

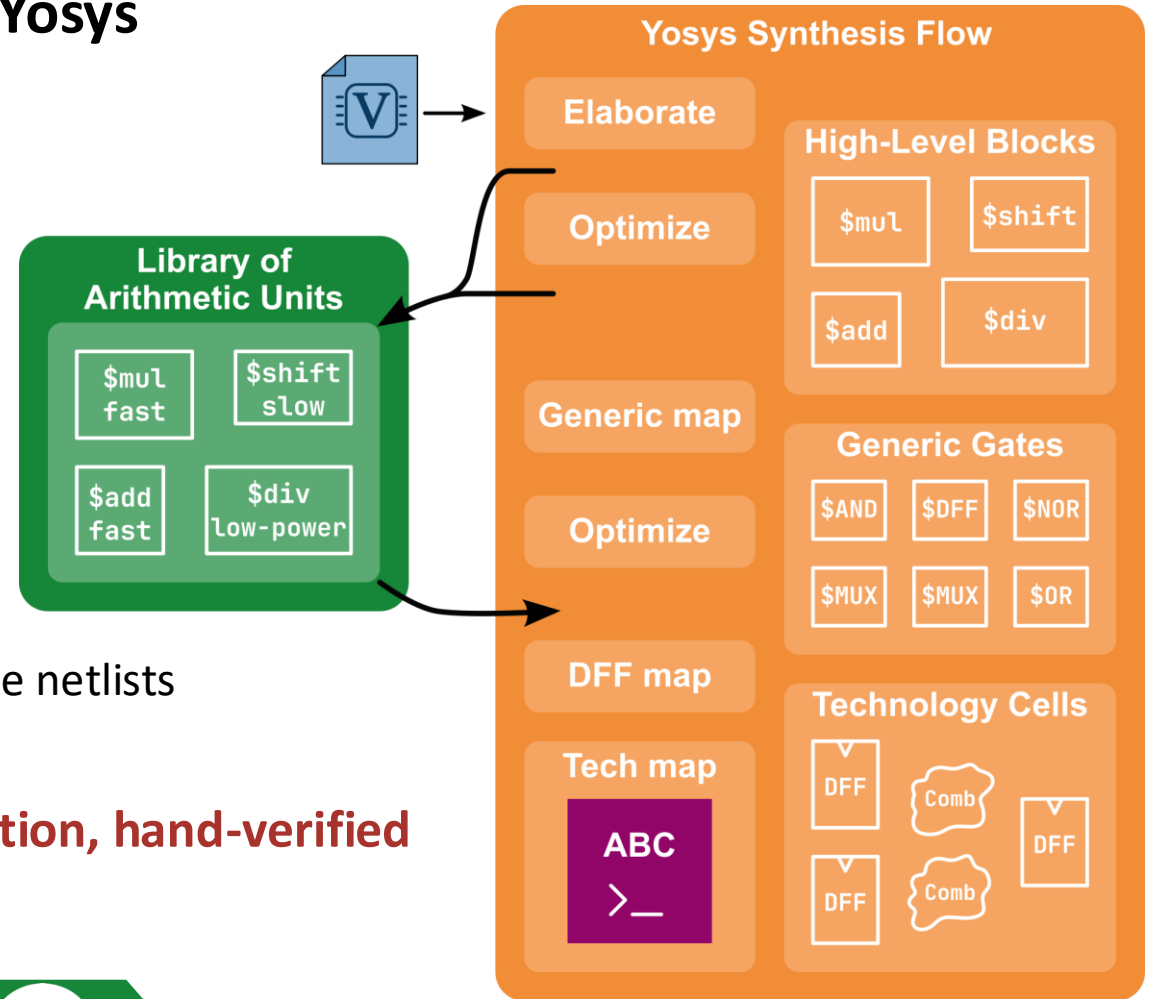


Optimized



Library of Arithmetic Unit (LAU)

- **Block replacement is implemented in Yosys**
 - Detect and replace arithmetic operators
 - Currently: manual selection
 - **Next: algorithmic, AI-based!**
- **No open-source LAU**
 - Rich, optimized library is key to good results
 - We built it
 - A wide range of arithmetic operations
 - 3 different performance variants of generic gate netlists
 - Thoroughly QoR evaluated and optimized
 - **SystemVerilog port from VHDL: LLM translation, hand-verified**



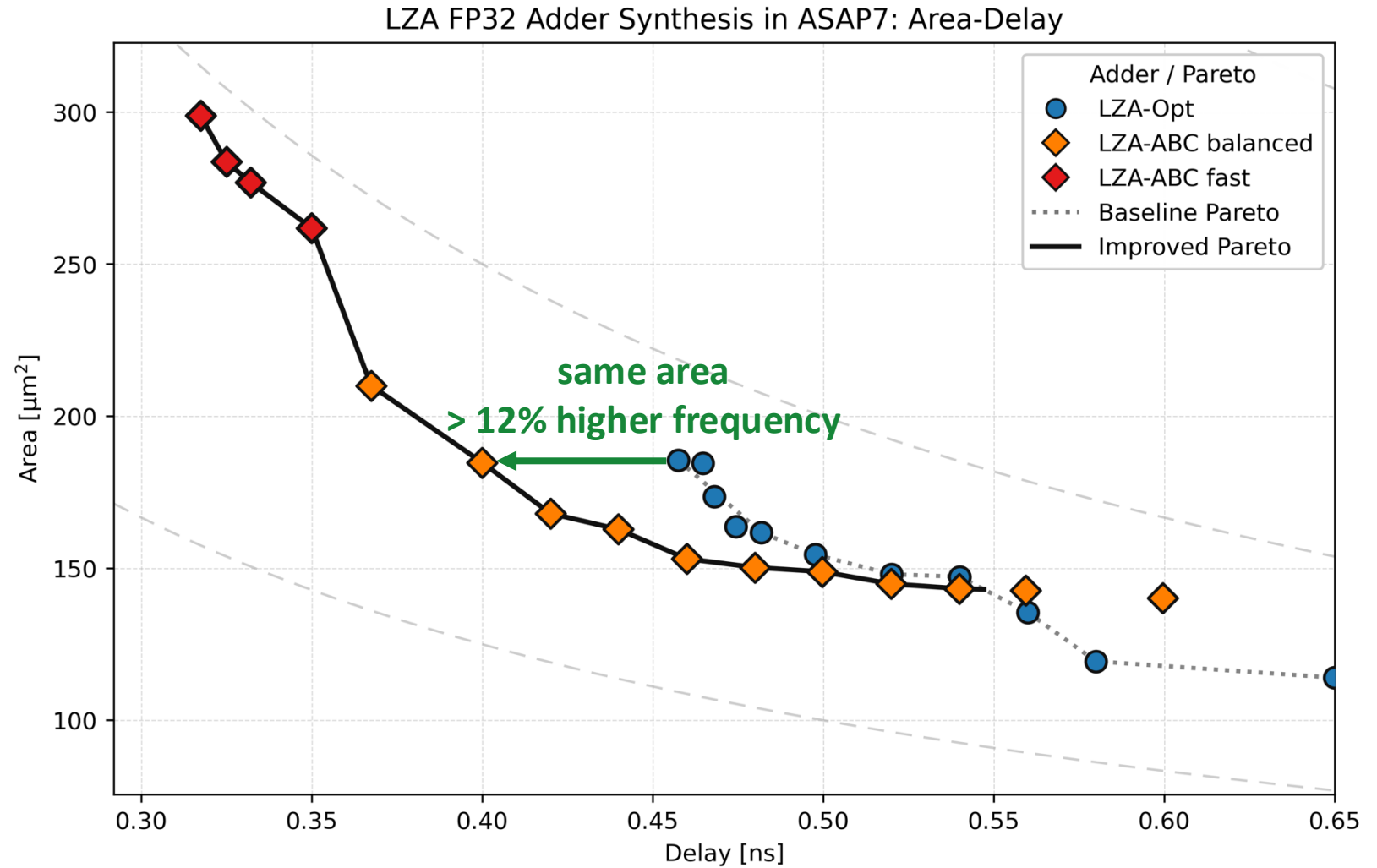
github.com/pulp-platform/elau



There is still room for improvement in Synthesis!



- Explored various FP32 adders:
 - Applied ABC logic optimizations before Synopsys Design Compiler synthesis, leading to **higher frequency** Pareto points for **several** designs.



End-to-end open designs need Open PDKs!



Booting Linux on

Basilisk

an end-to-end open-source
64-bit RISC-V SoC
in IHP 130nm BiCMOS



PULP



ETH zürich



• Open Design

- Cheshire with CVA6

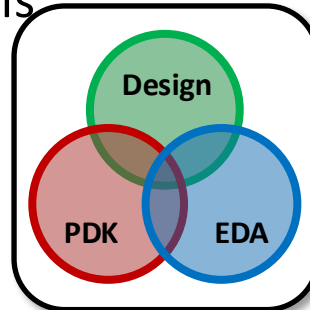
**More open PDKs
are needed!**

• OpenROAD (with Yosys (with slang))

- KiCAD (for board)

• Open PDK

- IHP 130
- Standard cells
- Memories



github.com/pulp-platform/cheshire-ihp130-o

At ETH Zürich, IC Design teaching uses open source HW

- **Starting 2025, our IC Design course switched to (mostly) open source**
 - Using IHP 130, Yosys and OpenROAD
 - Parts for backannotated simulation, test pattern generation, DRC/LVS, still use proprietary tools
- **Exercises based on a 32bit RISC-V microcontroller we call **Croc****

<https://github.com/pulp-platform/croc>

- **Project based grading**
 - Students (in groups of two) have to modify the exercise design
 - Best designs are taped out
- **Huge advantage:**
 - We can share lecture and exercise content with everyone

<https://vlsi.ethz.ch>

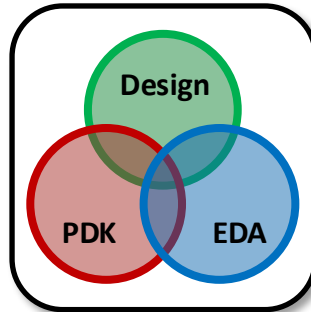


OpenROAD



YosysHQ

JYU

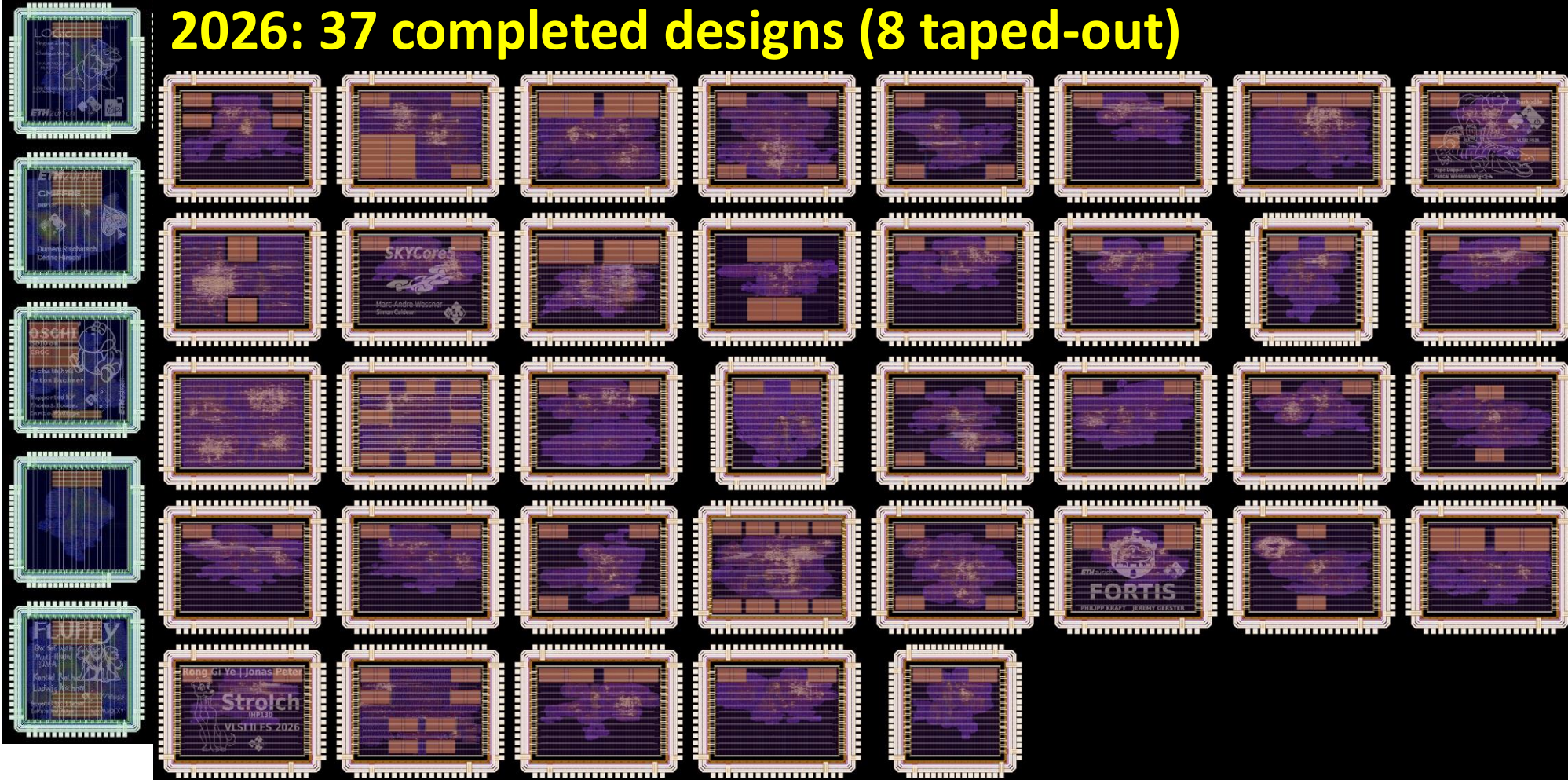


And the students delivered



2025: 33 completed designs (5 taped-out)

2026: 37 completed designs (8 taped-out)



Open Design Environments for European Chips: ODE4EC



- **A series of EU projects to support open-source EDA**
 - 20MEUR funding from EU, total project **50MEUR**
 - Organized in three sub projects: Digital, Analog, Productivity
 - Projects started 1st of June 2026
- **PULP is part of a large consortium**
 - 24 partners (DIG), 27 partners (AMS), 24 partners (PIV)
 - From 14 Countries (AT, DK, FI, FR, DE, GR, HU, IT, LT, PT, SI, ES, SE, SE, CH, UK)
 - Includes broad participation from most open source contributors in EU
- **Great opportunity to make a difference**
- **A fourth proposal is being prepared to support actual IC tape-outs**
 - Tentatively called ODE4EC-CES (Chips for Every Student)
 - 15MEUR funds to make 1000 student chips in Europe



Open Source HW is here to stay



- **Education and training**
 - The principles of IC Design remain the same between open and commercial tools
 - We have seen no decrease in learning quality when we switched to open-source EDA
- **Already good results and will only get better**
 - Especially with mature nodes, the PPA gap is much smaller than what people think
 - More work is needed for modern nodes, but we are getting there
 - Things will only get better
- **Innovation potential**
 - Being able to look into/influence the inner workings of EDA tools offers interesting innovation potential, especially for AI-based optimization pipelines.
- **The future is bright, and we are happy to be part of it**
 - Open-source SW changed the way we do SW. Why should open-source HW be different?